

The Theory of Locality-Sensitive Functions

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- LSH Families for Hamming Distance
- LSH Families for Cosine Similarity:
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Slides, Notes,

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Using Minhashing Directly:



Problem:

Even with compressed signatures, comparing all possible document pairs can be computationally expensive for large datasets.

Example:

With 1 million documents and 250-byte signatures, it would take about 6 days to compare all pairs, even if each comparison takes only a microsecond.

Solution:

Focus on pairs likely to be similar without examining every possible pair.

Method:

Locality-sensitive hashing (LSH) or near-neighbor search.

The Theory of Locality-Sensitive Functions:

- **Minhash functions** is one example of LSH family of functions that can be used to distinguish between pairs at a low distance from pairs at a high distance.
- **Low distance means similar --- High distance means dissimilar**
- Now, we shall explore other families of functions, besides the minhash functions, that can serve to produce candidate pairs efficiently.
- **Candidate pairs means similar pairs**
- There are three conditions that we need for a family of functions:

Three Conditions for a Function to be Part of Locality-Sensitive Functions Family:

1. They must be more likely to make close pairs be candidate pairs than distant pairs.
2. They must be statistically independent by giving a certain response by the product rule of independent events.
 - Two action movies might have an 80% chance of hashing to the same value.
 - An action movie and a drama might have a 20% chance.
3. They must be efficient, in two ways:
 - They must be able to identify candidate pairs in time much less than the time it takes to look at all pairs.
 - They must be combinable to build functions that are better at avoiding false positives and negatives, and the combined functions must also take time that is much less than the number of pairs.

Important Notations to Understand:

- we shall consider functions that take two items and make a candidate pair.
- If a function f will “hash” items, such that x and y becomes:
 - $f(x) = f(y)$
 - It means they should be a candidate pair.
 - We also use $f(x) \neq f(y)$ to mean “do not make x and y a candidate pair.
- This condition depends on distance between x and y :
- Let $d1 < d2$ be two distances according to some distance measure d . A family F of functions is said to be $(d1, d2, p1, p2)$ -sensitive if for every f in F :
 1. If $d(x, y) \leq d1$, then the probability that $f(x) = f(y)$ is at least $p1$.
 2. If $d(x, y) \geq d2$, then the probability that $f(x) = f(y)$ is at most $p2$.

Important Notations to Understand:

- 1. If $d(x, y) \leq d_1$, then the probability that $f(x) = f(y)$ is at least p_1 .
- 2. If $d(x, y) \geq d_2$, then the probability that $f(x) = f(y)$ is at most p_2 .

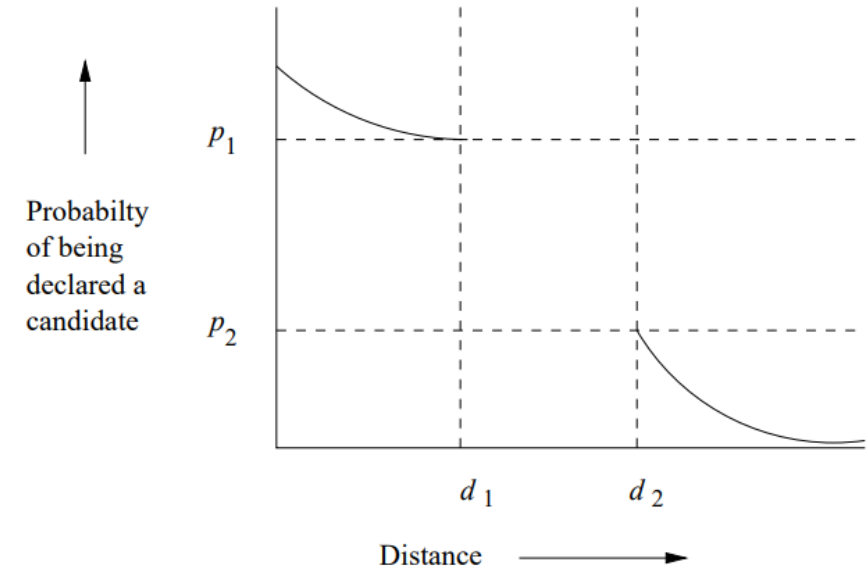


Figure 3.10: Behavior of a (d_1, d_2, p_1, p_2) -sensitive function

Minhash function with Jaccard distance:

- Using the family of minhash functions with the distance measure (Jaccard distance).
- if $d(x, y) \leq d1$ then Jaccard similarity $SIM(x,y) = 1 - d(x,y) \geq 1 - d1$
- The Jaccard similarity of x and y is equal to the probability that a minhash function will hash x and y to the same value
- Therefore, probability in this case is $\geq 1 - d1$
- $1 - d1$ represents the lower bound of the Jaccard similarity for items that are
- considered "close" or similar.
- Example:
 - Suppose sets A and B are given as: $A=\{1,2,3\}, B=\{2,3,4\}$
 - Their Jaccard similarity is:
 - $Jaccard\ similarity(A,B) = \frac{|A \cap B|}{|A \cup B|} = \frac{2}{4} = 0.5$
- After applying several permutations, Minhashing will produce a signature for A and B , and the fraction of times their Minhash values are equal will also be approximately 0.5.

LSH Families for Hamming Distance:

Locality-sensitive family of functions for Hamming distance.

1. Suppose we have a space of d -dimensional vectors, and $h(x, y)$ denotes the Hamming distance between vectors x and y .
2. If we take any one position of the vectors, say the i th position, we can define the function $f_i(x)$ to be the i th bit of vector x .
3. Then the probability that $f_i(x) = f_i(y)$ for a randomly chosen i is exactly $1 - h(x, y)/d$; i.e., it is the fraction of positions in which x and y agree.

Two key differences between Hamming distance & Jaccard distance :

1. Hamming distance ranges from 0 to d , while Jaccard distance ranges from 0 to 1.
2. The size of the family F for Hamming distance is limited to d , while minhash functions are essentially unlimited.

LSH Families for Cosine Similarity:

Cosine similarity is defined as the angle between how close two vectors are.

- In the context of locality-sensitive hashing (LSH) for cosine distance, A random hyperplane is chosen by picking a normal vector v .
- For two vectors x and y , if their dot products with v ($v \cdot x$ and $v \cdot y$) have the same sign, they're on the same side of the hyperplane.

- For example:
- If $v \cdot x$ and $v \cdot y$ are both positive or both negative, the vectors x and y are on the same side of the hyperplane.
- If one dot product is positive and the other is negative, the vectors are on opposite sides.

• **LSH Application:** In the LSH context, the probability of a randomly chosen hyperplane separating the two vectors is directly related to the angle θ between them.

- If the angle θ between two vectors is small (i.e., the vectors are similar), then the probability of them being separated by a random hyperplane is low, meaning they are more likely to have the same sign for their dot products with v .

- Conversely, if the angle is large (i.e., the vectors are dissimilar), the probability of them being separated by the hyperplane increases.

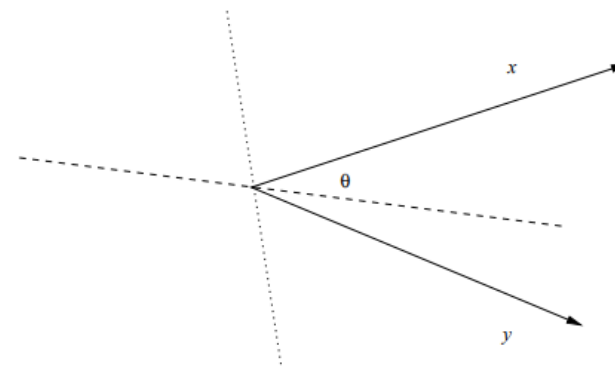


Figure 3.13: Two vectors make an angle θ

Locality-Sensitive Hashing (LSH) for Euclidean distance:

The Diagram:

- Shows a random line divided into segments of equal lengths ' a ' (called "Bucket width").
- Two points are projected onto this line, separated by a distance ' d '.
- An angle θ is shown, representing the angle between the line connecting the two points and the line onto which they're projected.

LSH for Euclidean Distance:

- Points are hashed to the bucket where their projection onto the line falls.

Probability of Hashing to the Same Bucket:

- If the distance ' d ' between two points is small compared to ' a ', they're likely to hash to the same bucket.
- For $d = a/2$, there's at least a 50% chance of hashing to the same bucket.
- The larger the angle θ , the higher the chance of hashing to the same bucket.

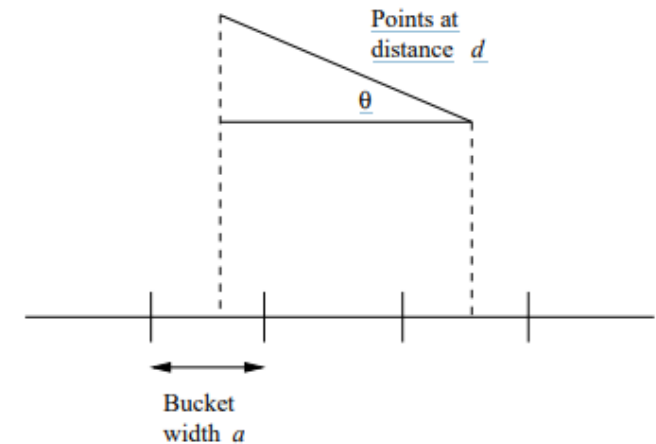


Figure 3.14: Two points at distance $d \gg a$ have a small chance of being hashed to the same bucket



Thank You